

Central Limit theorem

Sampling distribution,

will be normally distributed

$$x \approx N(\mu, \sigma)$$

$n = \text{Sample size}$

$$S_1 = \{x_1, x_2, \dots, x_n\} = \bar{x}_1$$

$$S_2 = \{x_1, \dots, x_n\} = \bar{x}_2$$

$$S_3 = \{x_1, \dots, x_n\} = \bar{x}_3$$

$$\vdots$$

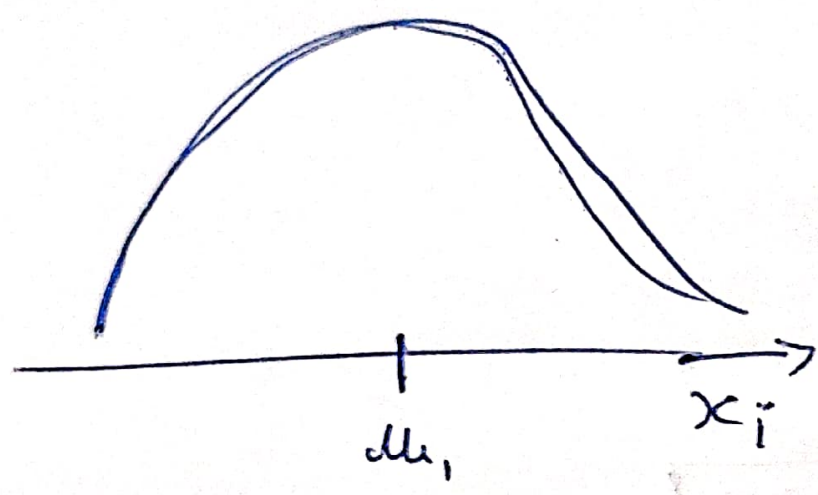
$$\vdots$$

$$\vdots$$

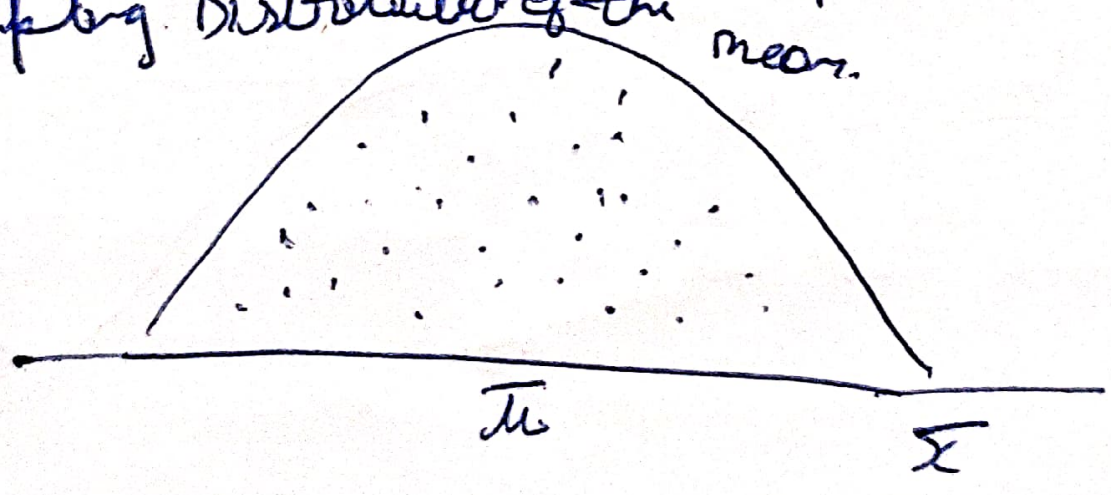
$$\vdots$$

$$\vdots$$

$$\vdots$$

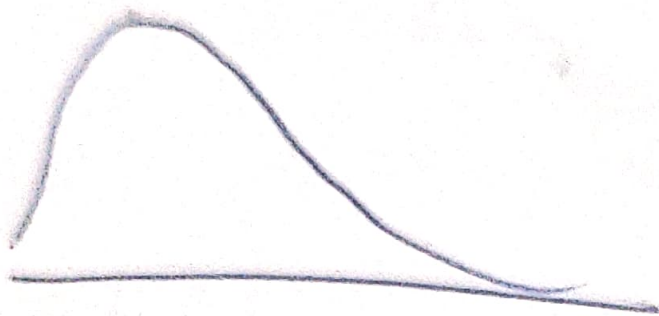


Sampling Distribution of the mean

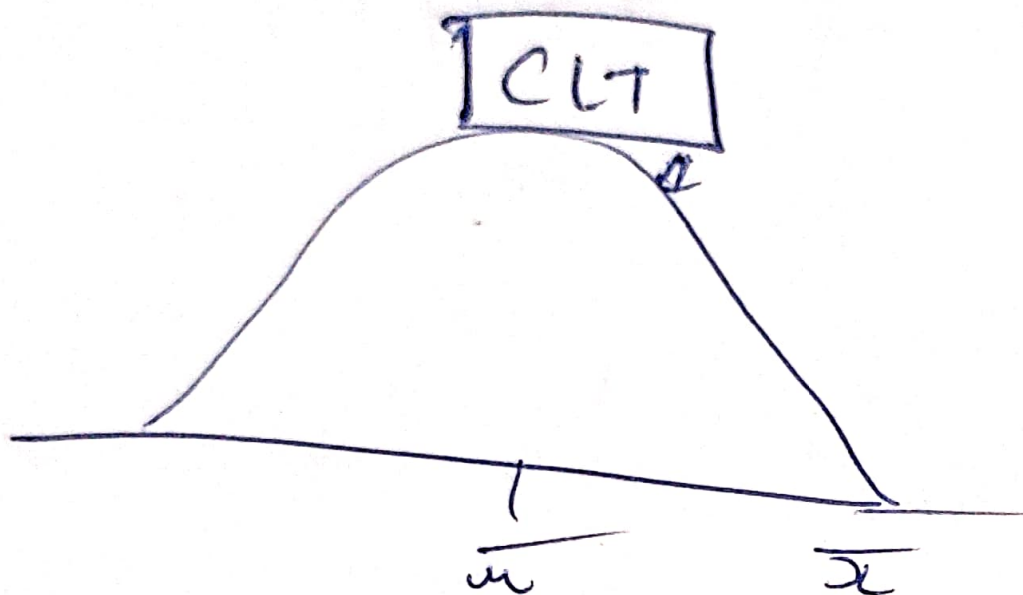


$$x \not\sim N(\mu, \sigma^2)$$

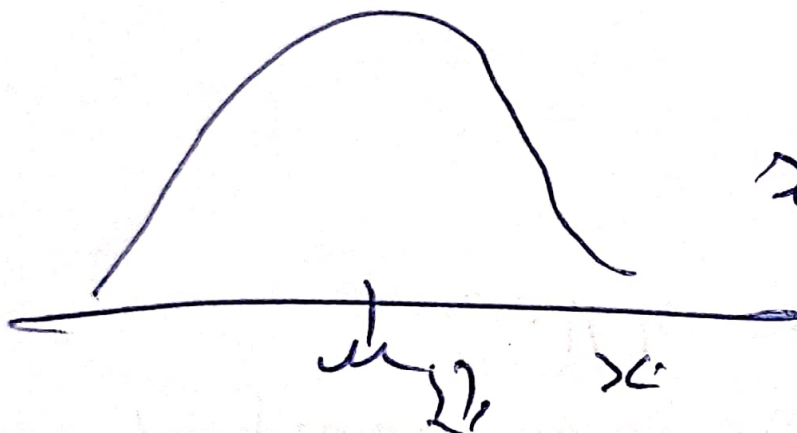
$$n \geq 30$$



$$\begin{matrix} S_1 & \bar{x}_1 \\ S_2 & \bar{x}_2 \end{matrix}$$

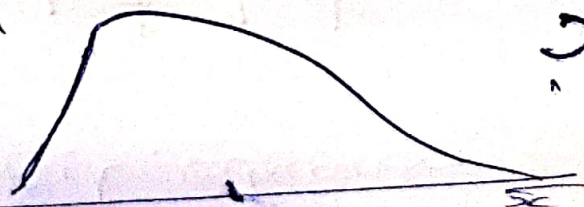


①



$$x \sim N(\mu, \sigma^2)$$

Sampling distribution



$$x \sim N(\mu, \sigma^2)$$